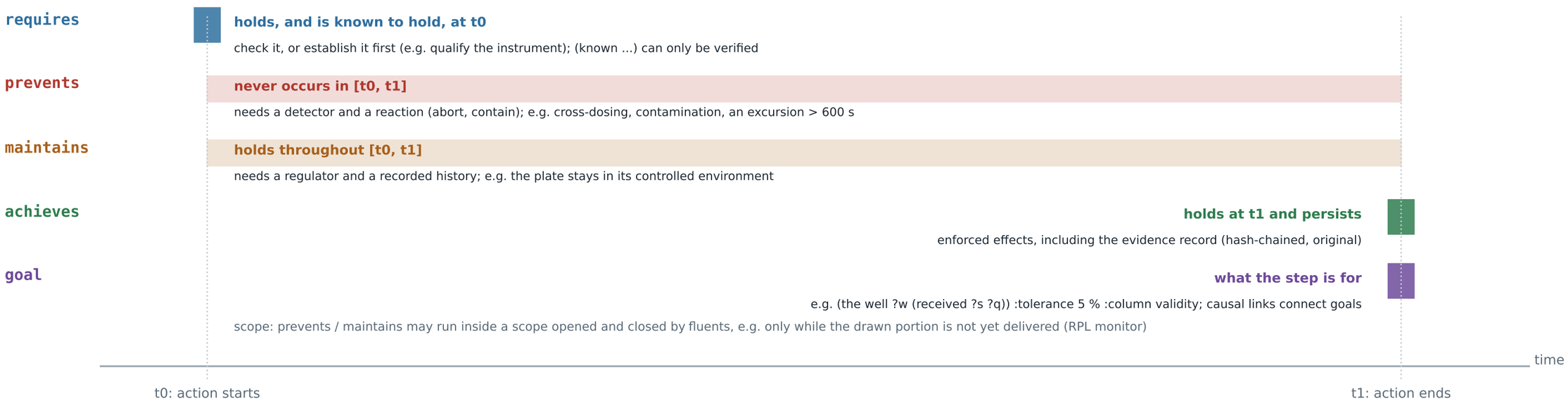
LDL: the lab designator language

states what a lab action must bring about - never how; every term is an entity description over lab entities, never a robot, tool, pose or device

A An action is a contract over its interval

`(the action ?a (type intervening) (into (the well ?w (of ?p))) (substance ?s) (dose ?q) ...)   - roles are bound by the step; the contract is owned by the action type`



instrument role (attributive only)

`(an instrument ?d (slot (a plate-slot (accepts ?p))) (process (a process (type substance-transfer) ...)))`

resolved by the realization to a device that has the slot and offers the process: dosing -> L12, imaging -> L17, incubation -> L10

signals and handles

`(signals (a failure (type dose-not-delivered)))  ->  (handles (on ...) (do ...) (then retry|replan|escalate))`

every handler states its guarantee: goal | goal-or-safe-state | safe-state | contained

B The lab's LDL documents, and what realizes them

`experiment.ldl` (72 lines)	the experiment as an entity: cultures defined by treatments, comparisons, goal (>= 0.5 bit), insights
`challenge_rescue.ldl` (112 lines)	the experiment's action designator: workflow at culture times, handlers with guarantees
`culture_plan.ldl` (83 lines)	culture-level plan: steps with physical and knowledge effects; order derived from causal links
`contracts.ldl` (95 lines)	category contracts: intervening, incubating, measuring, quarantining
`processes.ldl` (115 lines)	device processes: substance-transfer, imaging, incubation, containment; phases, irreversible phase, faults
`epistemic.ldl` (78 lines)	meta model: 10 rules, 8 treatments for beliefs that fall short (observed > reported > inferred > ...)
`experiment_concepts.ldl` (169 lines)	concept hierarchy of experiments with partonomy and closed-world checks

realized by

abstraction boundary (linted)

no robot, gripper, tool, pose, frame, prim, trajectory or device id;

instruments only attributively (a/an), never 'the'

rsa/ldl.py

parser, evaluator over belief and trail, admissibility: monitors for every prevents, a handler for every failure, every guarantee checked

rsa/ldl_to_cram.py, culture_to_cram.py

CRAM/RPL plans: partial order, with-policy monitors, unwind-protect, failure handling

kernel + robots (R01, R02, R03)

transport, tool coupling, sessions within the excursion window - the realization of one lab step

devices (L09, L10, L12, L17, L21, H4)

fill the instrument role through their slot and process; their records ground the process models

semantic digital twin (USD)

the referents: plates, wells, frames; records and belief

Checks (rsa/ldl.py, v0.3): 20/20 signalled failures handled, 13/13 repair guarantees hold, 11/11 record faults grounded, 12/12 negative controls caught. Quantities ILLUSTRATIVE (D016).